

QUESTION 1: ESSAY

An irreversible, reaction-limited, dual-site reaction $A + B \rightarrow C + D$ is taking place on a catalyst surface. When the P_D is doubled, the rate of reaction stays the same, while when P_C was doubled, the rate reduced. If, A and B both adsorb on the catalyst surface, write the most likely reaction rate equation.

Given Answer: Reaction limiting



$$r_{\text{overall}} = r_{\text{rxn}}$$

Notes:

- 1) Dual site => denominator squared
- 2) Numerator => adsorbed properties
- 3) P_D double = > No effect

Therefore,

$$r_{\text{overall}} = k (P_A P_B) / (1 + (k_A P_A) + (k_B P_B) + (k_C P_C))^2$$

Score: 25

QUESTION 2: ESSAY

(a) According to most correlations, what are the two **variables** that can change external mass transfer coefficient?

(b) How can the mass transfer resistance be reduced?



Given Answer: 1) Diffusivity

2) Boundary layer thickness

Mass transfer resistance is $1/k_c$

Want to increase k_c

=> Increase the flow rate by increasing stir rate

Score: 25

QUESTION 3: ESSAY

In a heterogeneous catalytic reaction, the measured order of reaction was different from the actual one. In fact, the $n_{\text{actual}} = 2 \cdot n_{\text{measured}} - 1$. Explain why the actual reaction-order was different.



Given Answer: Due to Falsified Kinetics

=> There is diffusion limitation present that skews the reaction order measurement.

Score: 20

QUESTION 4: ESSAY

For a reaction $A \rightarrow B$ on a catalyst surface, the process is mass-transfer limited. The concentration in the bulk is C_{Ab} and on the surface is C_{As} . The rate of surface reaction is given by $r_s = k_s C_{As}^2 / (1 + K_A C_{As})^2$. If the mass-transfer coefficient is k_c , what is the overall rate of reaction?



Given Answer: Mass Transfer limited

=> Series reactors

$r_{\text{overall}} = r_{\text{limiting}}$

$r_{\text{overall}} = r_{\text{MT}}$

$r_{\text{overall}} = k_c (C_{A_b} - C_{A_s})$

Score: 20